

Quantum Complexity in Position Space Scattering Simulations

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1 Models

1.1 1+1D Transverse Field Ising Model

- Free Hamiltonian in particle basis (PBC):

$$\hat{H}_0 = -J \sum_{i=1}^N \hat{\sigma}_i^x \hat{\sigma}_{i+1}^x - h_T \sum_{i=1}^N \hat{\sigma}_i^z.$$

- Interacting term:

$$\hat{H} = -g \sum_{i=1}^N \hat{\sigma}_i^z \hat{\sigma}_{i+1}^z,$$

turns the model into a XZ Heisenberg model.

- Three-fold diagonalization procedure:

- Jordan-Wigner Transformation:

$$\hat{c}_j^\dagger = \left(\prod_{i=1}^{j-1} \hat{\sigma}_i^z \right) \hat{\sigma}_j^-, \quad \hat{c}_j = \left(\prod_{i=1}^{j-1} \hat{\sigma}_i^z \right) \hat{\sigma}_j^+.$$

- Fourier Transformation:

$$\hat{\eta}_k^\dagger = \frac{1}{\sqrt{N_s}} \sum_{j=0}^{N-1} e^{-ikj} \hat{c}_j^\dagger, \quad \hat{\eta}_k = \frac{1}{\sqrt{N_s}} \sum_{j=0}^{N-1} e^{ikj} \hat{c}_j.$$

- Bogoliubov Transformation:

$$\hat{\xi}_k^\dagger = \cos \phi_k \hat{\eta}_k^\dagger + \sin \phi_k \hat{\eta}_{-k}, \quad \hat{\xi}_k = \cos \phi_k \hat{\eta}_k + \sin \phi_k \hat{\eta}_{-k}^\dagger.$$

(Inverse transform)

$$\hat{\eta}_k^\dagger = \cos \phi_k \hat{\xi}_k^\dagger - \sin \phi_k \hat{\xi}_{-k}, \quad \hat{\eta}_k = \cos \phi_k \hat{\xi}_k - \sin \phi_k \hat{\xi}_{-k}^\dagger.$$

such that

$$\tan \phi_k = \frac{J \sin k}{h_t - J \cos k}.$$

1.2 Gross-Neveu Model

- Free massive Hamiltonian (OBC) with L -sites and N -flavors:

$$\hat{H}_0 = \sum_{a=1}^N \left[\sum_{n=1}^{L-1} (\hat{\sigma}_{n,a}^y \hat{\sigma}_{n-1,a}^x - \hat{\sigma}_{n,a}^x \hat{\sigma}_{n+1,a}^y) + m \sum_{n=1}^L (-1)^n (\mathbb{1} - \hat{\sigma}_{n,a}^z) \right].$$

- Four-Fermi interaction term:

$$\hat{H}_G = \frac{G^2}{2} \sum_{n=1}^L \sum_{a=1}^N \sum_{b,b>a} (\mathbb{1} - \hat{\sigma}_{n,a}^z) (\mathbb{1} - \hat{\sigma}_{n,b}^z).$$

2 Quantum Complexity

- Bipartite entanglement entropy (subsystems A and B):

$$S(\rho_A) = -\text{Tr}(\rho_A \ln \rho_A).$$

– $\rho_A = \text{Tr}_B \rho$ is the reduced density matrix of ρ obtained by tracing out system B .

- Bipartite non-local magic:

$$M_2^{(\text{NL})}(\{\lambda_i\}) = -\log_2 \left(\sum_{i_1, i_2, i_3, i_4=0}^{r-1} \sqrt{\lambda_{i_1} \lambda_{i_2} \lambda_{i_3} \lambda_{i_4}} \times \sqrt{\lambda_{i_1 \wedge i_2 \wedge i_3} \lambda_{i_1 \wedge i_2 \wedge i_4} \lambda_{i_1 \wedge i_3 \wedge i_4} \lambda_{i_2 \wedge i_3 \wedge i_4}} \right).$$

- $\{\lambda_i\}$: singular values of the reduced density matrix ρ_A .
- $r = 2^{|A|(|B|)}$, where $|A|$ ($|B|$) is the size of bipartition $A(B)$ if $|A| \leq |B|$ ($|B| \leq |A|$).
- Bipartite non-local magic is the magic in the correlations between system A and B when basis contributions are minimized.

- Anti-flatness as a lower bound of bipartite non local magic:

$$\mathcal{F}(\rho_A) = \text{Tr} \rho_A^3 - (\text{Tr} \rho_A^2)^2.$$

3 Scattering State Preparation

Steps:

1. Construct ground state of the full interacting theory using DMRG or some other method.

There are tradeoffs for open and periodic boundary conditions. For periodic boundary conditions, you have translational invariance, allowing for well defined momenta.

3.1 1+1D Ising

- If the wave packets are far apart initially, then the interaction term is small enough that it can be ignored. So we can prepare wave packets in the ground state of the Ising model (free theory).
- Using

$$|k\rangle = \hat{\xi}_k^\dagger |\Omega\rangle,$$

we have

$$\hat{\eta}_k^\dagger |\Omega\rangle = (\cos \phi_k \hat{\xi}_k^\dagger - \sin \phi_k \hat{\xi}_{-k}^\dagger) |\Omega\rangle = \cos \phi_k |k\rangle,$$

so

$$|k\rangle = \frac{1}{\cos \phi_k} \hat{\eta}_k^\dagger |\Omega\rangle.$$

- Therefore we can define a wave packet in position space as

$$\begin{aligned} |\psi(k, x)\rangle &= \sum_{k'} e^{ik'x} e^{-|k'-k|^2/4\sigma_k^2} \hat{\xi}_k^\dagger |\Omega\rangle \\ &= \sum_{k'} e^{ik'x} e^{-|k'-k|^2/4\sigma_k^2} (\sec \phi_k \hat{\eta}_k^\dagger) |\Omega\rangle \\ &= \sum_{k'} e^{ik'x} e^{-|k'-k|^2/4\sigma_k^2} \sec \phi_{k'} \left(\frac{1}{\sqrt{N}} \sum_{x'} e^{-ik'x'} \hat{c}_{x'}^\dagger \right) |\Omega\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{x'} \left(\sum_{k'} e^{-ik'(x'-x)} e^{-|k'-k|^2/4\sigma_k^2} \sec \phi_{k'} \right) \hat{c}_{x'}^\dagger |\Omega\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{x'} \phi_{x'}^{k,x} \hat{c}_{x'}^\dagger |\Omega\rangle. \end{aligned}$$

Thus we define a Gaussian wave packet operator

$$\mathcal{G}^{k,x} = \frac{1}{\sqrt{N}} \sum_{x'} \phi_{x'}^{k,x} \hat{c}_{x'}^\dagger$$

for

$$\phi_{x'}^{k,x} = \sum_{k'} e^{-ik'(x'-x)} e^{-|k'-k|^2/4\sigma_k^2} \sec \phi_{k'}.$$

These can be built efficiently using the Givens rotation method.